

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1596

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 5

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I Dinesh V (192524309) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dinesh V

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks-I

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

The image displays three screenshots of the SIMATS Library Book Reservation System, a web application for library book management.

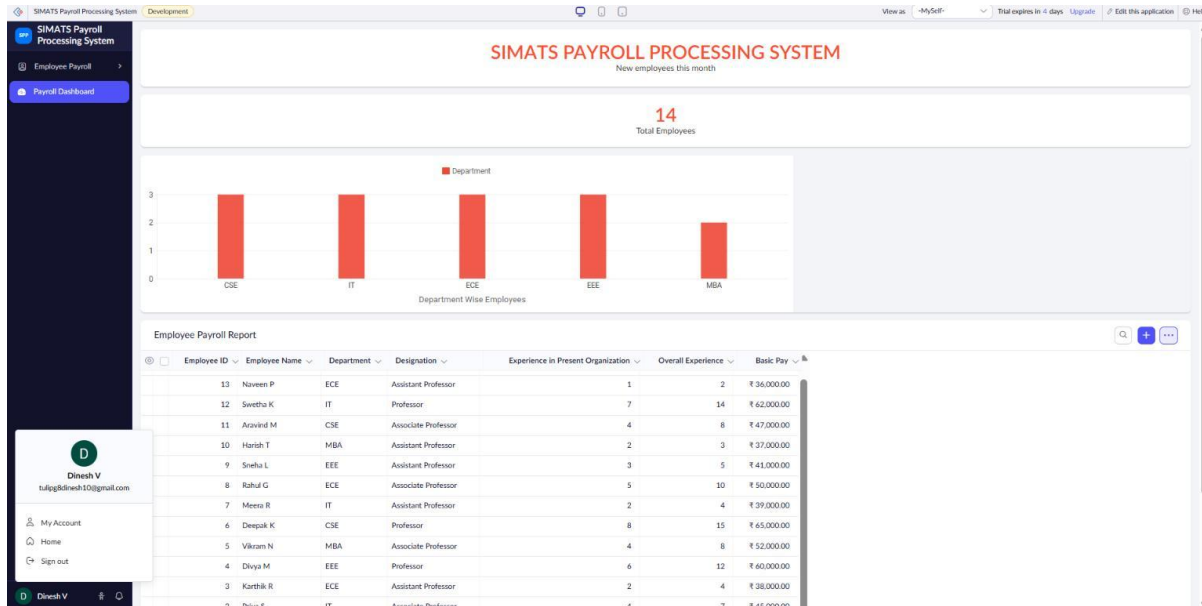
Dashboard Screenshot: The dashboard shows the system title "SIMATS Library Book Reservation System" and a subtitle "New employees this month". It features four summary cards: "20 Total Books", "14 Available Books", "4 Reserved Books", and "2 Returned Books". Below these is a pie chart representing the distribution of books by category: Computer Science, Mathematics, Physics, Chemistry, Electronics, Mechanical, Civil, and English. A "Library Books Report" table is also visible, showing columns for Book Id, Title of the Book, Author, Publication, Year, Edition, and Number of Copies.

Library Books Form Screenshot: This screenshot shows the form for adding a new book. The form includes fields for Title of the Book, Author, Publication, Year, Edition, Number of Copies, Rack Number, Book Category (a dropdown menu), Status (radio buttons for Available, Reserved, and Returned), Reserved By, and Reservation Date (dd-MMM-yyyy).

Library Books Report Screenshot: This screenshot shows the "Library Books Report" table, which lists the following data:

Book Id	Title of the Book	Author	Publication	Year	Edition	Number of Copies
20	Word Power Made Easy	Norman Lewis	Goyal Publishers	2021	1st	0
19	Organic Chemistry	Morrison & Boyd	Pearson	2017	7th	3
18	Data Communications & Networking	Berouz Forouzan	McGraw Hill	2022	5th	10
17	Surveying Vol 1	BC Punmia	Laxmi Publications	2019	16th	8
16	Fluid Mechanics	RK Bansal	Laxmi Publications	2020	10th	0
15	Digital Signal Processing	John Proakis	Pearson	2018	4th	4
14	Concepts of Physics	HC Verma	Bharati Bhawan	2021	1st	14
13	Discrete Mathematics	Kenneth Rosen	McGraw Hill	2019	8th	11
12	Artificial Intelligence	Stuart Russell	Pearson	2020	4th	0
11	Operating System Concepts	Silberschatz	Wiley	2021	10th	5
10	High School English Grammar	Wren & Martin	S. Chand	2022	120th	0
9	Structural Analysis	SS Bhavikatti	Vikas Publishing	2019	5th	7
8	Thermodynamics	PK Nag	McGraw Hill	2021	6th	9
7	Basic Electronics Engineering	BL Theraja	S. Chand	2018	23rd	0
6	Engineering Chemistry	Jain & Jain	Dhanpat Rai	2020	16th	6

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.



The screenshot displays the SIMATS Payroll Processing System Employee Payroll form. The form includes fields for Employee Name, Department (a dropdown menu), Designation, Experience in Present Organization, Overall Experience, Basic Pay, HRA, DA, and TA. Each of the last five fields (Basic Pay, HRA, DA, TA) has a mask of six hash symbols (#####). A Submit button is located at the bottom of the form. The left sidebar shows the navigation menu with options: Employee Payroll, Employee Payroll Report, All Employees, CSE Employees, payroll, and Payroll Dashboard. The user profile at the bottom left shows 'Dinesh V' with the email 'tulipg8dinesh10@gmail.com'.

SIMATS Payroll Processing System

Development

View as -MySelf- Trial expires in 4 days Upgrade Edit this application Help

SIMATS Payroll Processing System

Employee Payroll

Employee Payroll

Employee Payroll Report

All Employees

CSE Employees

payroll

Payroll Dashboard

D

Dinesh V

tulpg8dinesh10@gmail.com

My Account

Home

Sign out

Employee Payroll Report

Employee ID	Employee Name	Department	Designation	Experience in Present Organization	Overall Experience	Basic Pay
14	Kavya D	EEE	Associate Professor	5	9	₹ 49,000.00
13	Naveen P	ECE	Assistant Professor	1	2	₹ 36,000.00
12	Swetha K	IT	Professor	7	14	₹ 62,000.00
11	Aravind M	CSE	Associate Professor	4	8	₹ 47,000.00
10	Harish T	MBA	Assistant Professor	2	3	₹ 37,000.00
9	Sneha L	EEE	Assistant Professor	3	5	₹ 41,000.00
8	Rahul G	ECE	Associate Professor	5	10	₹ 50,000.00
7	Meera R	IT	Assistant Professor	2	4	₹ 39,000.00
6	Deepak K	CSE	Professor	8	15	₹ 65,000.00
5	Vikram N	MBA	Associate Professor	4	8	₹ 52,000.00
4	Divya M	EEE	Professor	6	12	₹ 60,000.00
3	Karthik R	ECE	Assistant Professor	2	4	₹ 38,000.00
2	Priya S	IT	Associate Professor	4	7	₹ 45,000.00
1	Arun Kumar	CSE	Assistant Professor	3	5	₹ 40,000.00

Showing 14 of 14

SIMATS Payroll Processing System

Development

View as -MySelf- Trial expires in 4 days Upgrade Edit this application Help

SIMATS Payroll Processing System

Employee Payroll

Employee Payroll

Employee Payroll Report

All Employees

CSE Employees

payroll

Payroll Dashboard

D

Dinesh V

tulpg8dinesh10@gmail.com

My Account

Home

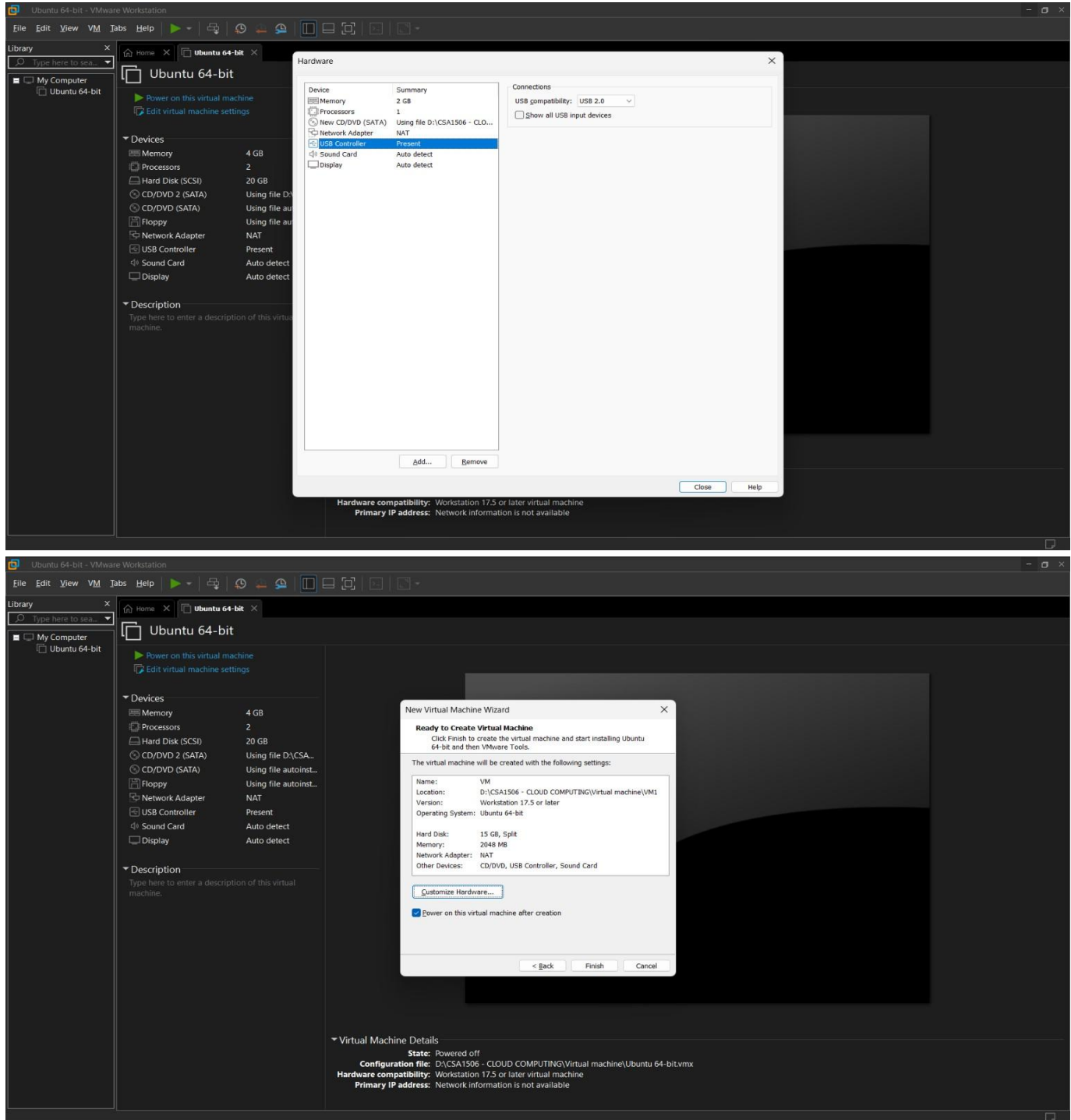
Sign out

All Employees

Employee ID	Employee Name	Department	Designation	Experience in Present Organization	Overall Experience	Basic Pay
14	Kavya D	EEE	Associate Professor	5	9	₹ 49,000.00
13	Naveen P	ECE	Assistant Professor	1	2	₹ 36,000.00
12	Swetha K	IT	Professor	7	14	₹ 62,000.00
11	Aravind M	CSE	Associate Professor	4	8	₹ 47,000.00
10	Harish T	MBA	Assistant Professor	2	3	₹ 37,000.00
9	Sneha L	EEE	Assistant Professor	3	5	₹ 41,000.00
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6	Deepak K	CSE	Professor	8	15	₹ 65,000.00
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4	Divya M	EEE	Professor	6	12	₹ 60,000.00
3	Karthik R	ECE	Assistant Professor	2	4	₹ 38,000.00
2	Priya S	IT	Associate Professor	4	7	₹ 45,000.00
1	Arun Kumar	CSE	Assistant Professor	3	5	₹ 40,000.00

Showing 14 of 14

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.



Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

